

Foundations of Responsible AI

Generative AI

Module 2 – Lesson 2

Today's activities

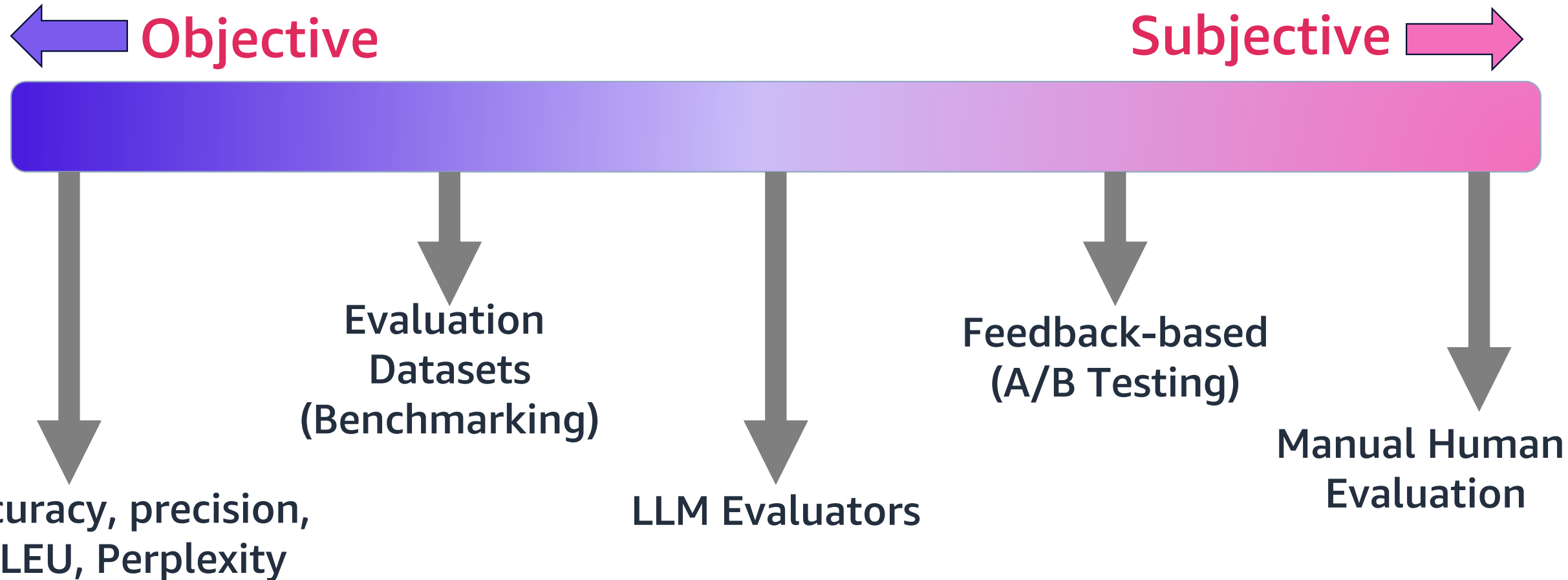
- Review of lesson 1
- Introduction to responsible AI
- Assessing the risk of an AI application



Review: Why is it important to evaluate LLMs?

- Increasing number of core abilities
- Knowledge of diverse topics
- Increasing visibility and influence on society
- Use cases beyond content generation
- Experts for other systems

Review: Evaluating LLMs



Introduction to responsible AI

What is responsible AI?

“ Responsible AI is the practice of **designing, developing, and using** AI technology with the goal of maximizing benefits and minimizing risks and unintended harms.

In practice, responsible AI is defined using a core set of dimensions that are assessed and updated over time as technology evolves.

Organizational culture of responsible AI

- RAI is an **organizational structure**, a set of principles (**dimensions**), and **practice**
 - Responsible AI is not a subdomain of AI
- There are several different dimensions of responsible AI; each dimension depends on organization and RAI maturity
- New RAI dimensions can emerge as new scientific evidence comes to light

Why so much focus on responsible AI?

- Building and maintaining **customer trust** is a key priority
- **Unwanted** model output includes
 - Hallucinations, inaccuracies
 - Instructions to share private information
 - Biased, aggressive, hateful, or demeaning text
 - Lawless text or unlicensed content
 - ...
- Many **new risks** have arisen with generative AI
 - Customer obsession drives us to provide safe and accurate models

Regulatory landscape under development

- Various standards, recommendations, and regulations are currently being discussed to better understand and mitigate risk of AI systems



Responsible AI dimensions

Controllability

Having mechanisms to monitor and steer AI system behavior

Privacy & Security

Appropriately obtaining, using and protecting data and models

Safety

Preventing harmful system output and misuse

Fairness

Considering impacts on different groups of stakeholders

Veracity & Robustness

Achieving correct system outputs, even with unexpected or adversarial inputs

Explainability

Understanding and evaluating outputs generated by an AI system

Transparency

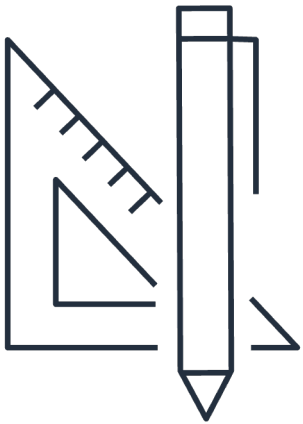
Enabling stakeholders to make informed choices about their engagement with an AI system

Governance

Incorporating best practices into the AI supply chain, including providers and deployers

Who should think about RAI?

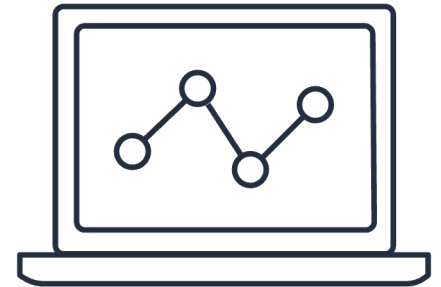
Design



Build



Operate



Everyone must consider RAI!

Design

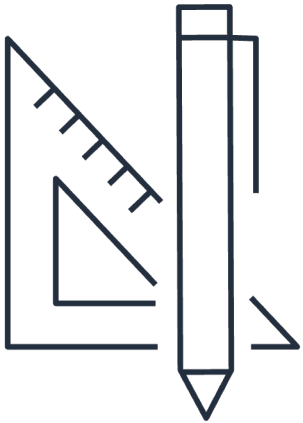
Build

Operate

No matter which part of the lifecycle
you're working on, you should always
consider RAI

Define business problem and assess risk

Design



- Discuss the use case with different stakeholders and **diverse groups**.
- Evaluate if AI adds **value** to any potential solution.
- Conduct a thorough **risk assessment** of the proposed use case.

Train, test and evaluate an AI/ML system

- Verify training data is **safe**, **relevant**, and **representative**.
- Consider **legal requirements** where applicable (e.g. licensing, privacy, consent).

Build

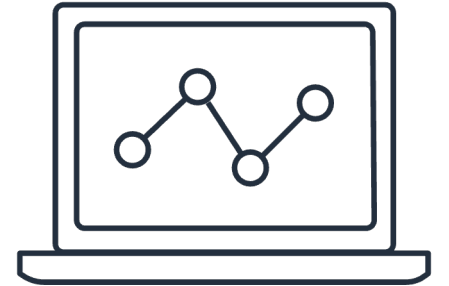


- ⚙️ Use **metrics** to evaluate outcomes; **include confidence intervals**.
- ⚙️ Consider model disgorgement, **value alignment**, and use **safeguards** to mitigate risk.

Deploying and using an AI/ML system

- Offer end users a way to **inquire about outputs** for high risk use cases (through e.g. feedback/request forms) and **be transparent about limitations**.
- Check for **model drift** (the world evolves) and **ensure the model is used as intended** (e.g. trained on US, should be used for US).

Operate



Assessing the risk of an AI application

Risk assessment process



Define the use case and
relevant stakeholders

Risk assessment process

1

Define the use case and
relevant stakeholders

2

Identify harmful events,
evaluate inherent and
residual risk



Risk assessment process

1

Define the use case and relevant stakeholders

2

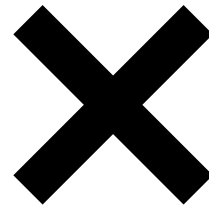
Identify harmful events, evaluate inherent and residual risk

3

Summarize risk levels for all risk dimensions and conclude findings

Quantifying AI system risk

measure of an event's
probability of occurring



magnitude or degree of
the consequences

likelihood

severity

Source: National Institute of Standards and Technology, [AI Risk Management Framework 1.0](#)

Let's summarize risk for one dimension

Veracity (risk)

Generating content that is not verifiably correct
(such as hallucinations from generative AI
systems).

Likelihood scale and criteria

Likelihood scale	Criteria
Frequent:	Occurs more than 100 times a year; or probability is close to 100%.
Likely:	Occurs between 10-100 times a year; or probability is between 50% to 99%.
Possible:	Occurs between 1-10 times a year; or probability is between 1% to 49%.
Unlikely:	Occurs less than 1x/year, but > once every 10 years; or probability is close to 1%.
Highly unlikely:	Occurs less than once every 10 years; or probability is nearly 0%.

Severity scale example: Veracity

Severity scale	Criteria
Extreme:	Persuasive, dangerous output results in irreversible negative outcome to the real world, including social impact.
Major:	Largely fabricated output results in substantial negative impact to the real world, including social impact.
Moderate:	Hallucinated output that leads to clear errors and degrades model performance and reliability in key tasks.
Low:	Hallucinations output causes minor errors but does not significantly alter model performance.
Very low:	Minimal amount of hallucination in output that causes negligible impact on model performance.

Likelihood and severity

Severity

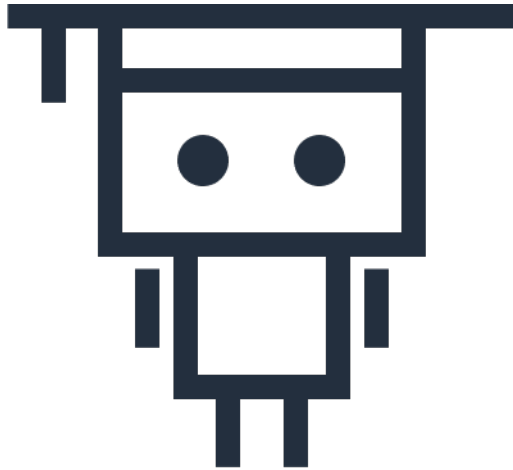
		Very low	Low	Moderate	Major	Extreme
Likelihood	Frequent	Low	Medium	High	Critical	Critical
	Likely	Very Low	Low	Medium	High	Critical
	Possible	Very Low	Low	Medium	High	Critical
	Unlikely	Very Low	Very Low	Low	Medium	High
	Highly unlikely	Very Low	Very Low	Very Low	Low	High

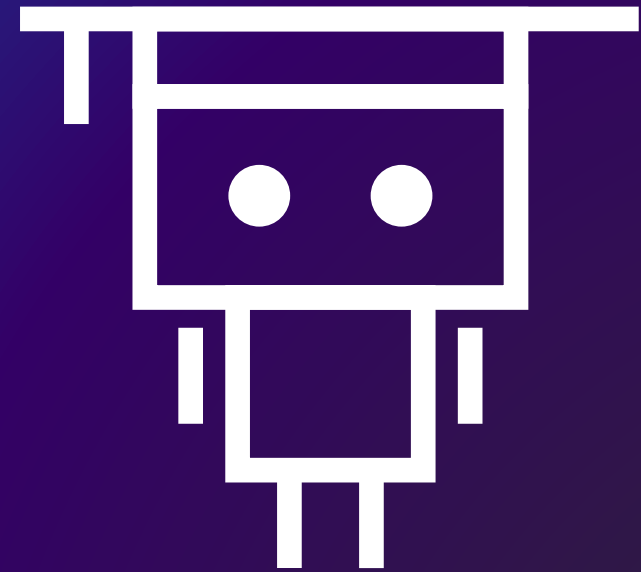
Key takeaways

- Designing, building and operating AI responsibly can help mitigate harm, build trust with customers, and allows organizations align with relevant regulatory requirements.
- Responsible AI has different dimensions; as the science evolves dimensions may change.
- Governance helps to carry out RAI practices across teams and organizations.

Next lesson

- This lesson covered topics related to responsible AI.
- In the next lesson, you will learn about specific dimensions of responsible AI.





Thank you!